Pseudocode for an ABM of Brokerage in Matching Markets

1 State, Notation, and Matching Function

Blue notation, equations, and algorithm lines are diagnostics-only measures. They are recorded for $\mathcal{M}$ or downstream figures, exploration scripts, and analyses; they do not enter transition, choice, learning, or matching rules.

For any matrix Z, $Z_{\cdot\ell}$ denotes column ℓ and $Z_{\cdot L}$ denotes the submatrix with ordered column-index set L. For a vector z, z_L denotes the corresponding subvector.

Table 1: Cross-routine symbols and dimensions.

Symbol	Dimension/domain	Definition
Core		
N	scalar	Number of agents, with agent set $[N] = \{1, \dots, N\}$.
ϑ	parameter vector	Simulation parameter vector.
seed	integer	Random seed.
S^t	state tuple	Mutable simulation state at end of period t.
T	integer	Number of simulation periods.
$\mathcal{M} = [m^1; \dots; m^T]$	$\mathbb{R}^{T \times m}$	Measure table; row $m^t = \mathcal{M}_{t\cdot} \in \mathbb{R}^m$.
Parameters		
d	scalar	Agent type dimension.
k_G	scalar	Watts-Strogatz graph degree.
ω	scalar	Satisfaction EWMA weight.
E_{init}	scalar	Initial training steps.
E_{steps}	scalar	Minimum (floor) training steps per period.
η_{lr}	scalar	MLP Adam learning rate.
W_p	scalar	Training-window horizon, in periods.
M	scalar	Per-call training observation cap.
Agent types, networks, and output		
$G^t = (V, \mathcal{E}_G^t)$	graph	Undirected graph, $V = [N] \cup \{N+1\}$; node $N+1$ is the broker.
$X^t = [x_1^t, \dots, x_N^t]$	$\mathbb{R}^{d \times N}$	Agent type matrix; $x_i^t = X_{\cdot i}^t \in \mathbb{R}^d$.
γ	$[0,1] \to \mathbb{R}^d$	Agent type curve for initial and entrant draws.
q_{ij}^t	scalar	Realized output for match (i, j).
True matching function		
Q	scalar	Output offset.
ρ	scalar	Quality-interaction weight.
δ	scalar	Regime-gain amplitude.
σ_ε	scalar	Output-noise standard deviation.
c	$\mathbb{R}^d$	Ideal agent type.
A	$\mathbb{R}^{d \times d}$	Symmetric positive definite interaction matrix.
B	$\mathbb{R}^{d \times d}$	Symmetric regime operator.
Calibration and costs		

Symbol	Dimension/domain	Definition
$\bar{q}_{\mathrm{cal}}$	scalar	Calibration mean.
r	scalar	Reservation output.
ϕ	scalar	Broker fee.
c_s	scalar	Self-search cost.
Learning and predictions		
d_b	integer	Broker symmetric pair-feature dimension, $d + d(d+1)/2$.
$\psi_b(x_i, x_j)$	$\mathbb{R}^{d_b}$	Symmetric broker pair feature vector.
$\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$	$\Xi_i^t \in \mathbb{R}^{d \times n_i^t}$, $y_i^t \in \mathbb{R}^{n_i^t}$	Agent i history; observation ℓ is partner agent type $\Xi_{i,\cdot\ell}^t$ and output $y_{i\ell}^t$.
$\mathcal{H}_b^t = (X_{b,1}^t, X_{b,2}^t, y_b^t, n_b^t)$	$X_{b,1}^t, X_{b,2}^t \in \mathbb{R}^{d \times n_b^t}$, $y_b^t \in \mathbb{R}^{n_b^t}$	Broker history; observation ℓ stores the two party types and output $y_{b\ell}^t$.
Θ_i^t	NN parameters	Agent i MLP weights.
Θ_b^t	NN parameters	Broker MLP weights.
$\mathcal{O}_i^t, \mathcal{O}_b^t$	Adam states	Persistent optimizer moments and step counters.
Δ_i^t, Δ_b^t	nonnegative integers	Observations added since the learner's previous training call.
$\mathbf{p}_i^t, \mathbf{p}_b^t$	integer sequences	Cumulative history counts at the end of learner initialization (period 0) and each later completed period.
$v_b^t(i, j)$	scalar	Symmetric broker prediction for unordered pair $\{i, j\}$.
$w_{j\leftarrow i}^t$	scalar	Receiver acceptance score.
$\bar{q}_{i\cdot}^t$	vector	Agent i's partner-indexed empirical means.
$C_{i\cdot}^t$	vector	Agent i's partner-indexed counts.
Period state and market objects		
s_i^t	$\mathbb{R}^2$	Satisfaction scores for self and broker channels.
χ_i^t	binary	Indicator that agent i has ever used the broker.
M_i^t	set	Agent i current-period counterparties.
z_i^t	channel	Channel chosen by demander i.
d_i^t	integer	Current-period demand.
D^t	subset of $[N]$	Current broker clients, $\{i : d_i^t > 0, z_i^t = \text{broker}\}$.
$\mathcal{A}_b^t$	subset of $[N]$	Broker access set, $\text{Roster}^t \cup D^t$.
Roster^t	subset of $[N]$	Standing roster.
rep_b^t	scalar	Broker reputation.
D_+^t	subset of $[N]$	Broker clients with positive residual offer quota.
$\mathcal{D}^t$	list	Demand records (i, z_i^t, d_i^t) with $d_i^t > 0$.
$\mathcal{R}^t$	list	Accepted records $(i, j, z, q, \hat{q}, o_1, o_2)$.
rng^t	RNG state	State of the simulation random-number stream.

The mutable simulation state at the end of period t is

$$S^t = \begin{pmatrix} \vartheta, \mathsf{rng}^t, \gamma, c, A, B, \bar{q}_{\mathrm{cal}}, r, \phi, c_s, X^t, G^t, \\ \{\mathcal{H}_i^t, \Theta_i^t, \mathcal{O}_i^t, \Delta_i^t, \mathbf{p}_i^t, \bar{q}_{i\cdot}^t, C_{i\cdot}^t, s_i^t, \chi_i^t, M_i^t\}_{i=1}^N, \\ \mathcal{H}_b^t, \Theta_b^t, \mathcal{O}_b^t, \Delta_b^t, \mathbf{p}_b^t, \text{Roster}^t, D^t, \text{rep}_b^t \end{pmatrix}.$$

This tuple includes fixed realized model objects and all variables needed for later transitions. Implementation-only scratch buffers and diagnostic-only counters are omitted.

The matching function is

$$f(x_i, x_j) = \rho \, \frac{x_i^\top c + x_j^\top c}{2} + (1-\rho)\, g(x_i, x_j)\, x_i^\top A x_j, \qquad g(x_i, x_j) = 1 + \delta \, \mathrm{sign}(x_i^\top B x_j).$$

2 Top-Level Simulation

Algorithm 1 Brokerage Simulation

Require: Parameters ϑ and seed.
Ensure: Final state S^T and measure table $\mathcal{M} \in \mathbb{R}^{T \times m}$.

1: $S^0 \leftarrow$ Initialize(ϑ, seed)
2: **for** $t = 1, \ldots, T$ **do**
3: $\quad (S^t, \mathcal{R}^t, Y^t) \leftarrow$ PeriodUpdate(S^{t-1}, t, ϑ)
4: $\quad \mathcal{M}_{t\cdot} \leftarrow Y^t$
5: **return** $(S^T, \mathcal{M})$

3 Training

For any one-hidden-layer predictor with input dimension p_{in} and hidden width h,

$$\mathrm{NN}_\Theta(\xi) = w_2^\top \,\mathrm{ReLU}(W_1 \xi + b_1) + b_2, \qquad \Theta = (W_1, b_1, w_2, b_2),$$

where $\xi \in \mathbb{R}^{p_{\mathrm{in}}}$, $W_1 \in \mathbb{R}^{h \times p_{\mathrm{in}}}$, $b_1, w_2 \in \mathbb{R}^h$, and $b_2 \in \mathbb{R}$.

$$d_b = d + \frac{d(d+1)}{2}, \qquad \psi_b(x_i, x_j) = \left[x_i + x_j;\ \mathrm{vech}\left(\frac{x_i x_j^\top + x_j x_i^\top}{2} \right) \right].$$

Agent networks use $(p_{\mathrm{in}}, h) = (d, 2d)$; the broker network uses $(p_{\mathrm{in}}, h) = (d_b, 8d)$.

$$\psi_b(x_i, x_j) = \psi_b(x_j, x_i)$$

so broker histories produce one training column per observed unordered pair.

Local variables for training

Symbol	Dimension/domain	Definition
p_{in}, h	integers	Predictor input dimension and hidden width.
ξ	$\mathbb{R}^{p_{\mathrm{in}}}$	Predictor input vector.
Θ	NN parameters	Generic MLP parameters (W_1, b_1, w_2, b_2).
W_1, b_1, w_2, b_2	matrix, vectors, scalar	One-hidden-layer MLP weights and biases.
d_b	integer	Broker symmetric pair-feature dimension.
ψ_b	function	Symmetric broker pair-feature map.
$\Xi_{\mathrm{tr}}, y_{\mathrm{tr}}$	$\mathbb{R}^{p_{\mathrm{in}} \times n}$, $\mathbb{R}^n$	Generic training inputs and targets.
$\mathcal{L}$	scalar	Full-batch MSE objective.
e	integer	Update-step index.
(m, v, τ)	Adam state	Persistent Adam moments and step counter; $\mathcal{O}_i, \mathcal{O}_b$ are the agent and broker instances.
$\beta_1, \beta_2, \epsilon$	scalars	Adam constants (0.9, 0.999, 10^{-8}).

Local variables for training (continued)		
Symbol	Dimension/domain	Definition
n_i, n_b	integers	Current values of history lengths n_i^t and n_b^t.
L_i, L_b	ordered index sets	Training-window indices from the most recent W_p completed simulation periods, with period 0 included only until W_p later periods exist; subsampled evenly to at most M.
Δ_i, Δ_b	integers	New observations since last training call.
E_i, E_b	integers	Agent and broker update-step counts.
$\Xi_{i,\mathrm{tr}}, y_{i,\mathrm{tr}}$	$\mathbb{R}^{d\times\lvert L_i\rvert}$, $\mathbb{R}^{\lvert L_i\rvert}$	Agent i training view of $\mathcal{H}_i^t$.
$\Psi_{b,\mathrm{tr}}, y_{b,\mathrm{tr}}$	$\mathbb{R}^{d_b\times\lvert L_b\rvert}$, $\mathbb{R}^{\lvert L_b\rvert}$	Broker symmetric-feature training view of $\mathcal{H}_b^t$.

Algorithm 2 TrainNN

Require: Network parameters Θ, inputs $\Xi_{\mathrm{tr}} \in \mathbb{R}^{p_{\mathrm{in}}\times n}$, targets $y_{\mathrm{tr}} \in \mathbb{R}^n$, steps E, learning rate η_{lr}, persistent Adam state (m, v, τ).

for $e = 1, \ldots, E$ **do**
  $g \leftarrow \nabla_\Theta \mathcal{L}(\Theta), \quad \mathcal{L}(\Theta) = n^{-1}\sum_{\ell=1}^{n}\left(\mathrm{NN}_\Theta(\Xi_{\mathrm{tr},\cdot\ell}) - y_{\mathrm{tr},\ell}\right)^2$
  $\tau \leftarrow \tau + 1$
  $m \leftarrow \beta_1 m + (1-\beta_1)g, \quad v \leftarrow \beta_2 v + (1-\beta_2)\, g \odot g$
  $\Theta \leftarrow \Theta - \eta_{\mathrm{lr}}\,\dfrac{m/(1-\beta_1^\tau)}{\sqrt{v/(1-\beta_2^\tau)} + \epsilon}$
return $\Theta, (m, v, \tau)$

Within period-level routines, state variables without a superscript refer to their current-period values.

Algorithm 3 TrainPredictors

Require: State S, period t, parameters $(E_{\mathrm{init}}, E_{\mathrm{steps}}, \eta_{\mathrm{lr}}, W_p, M)$.

for $i = 1, \ldots, N$ in parallel **do**
  $n_i \leftarrow n_i^t$, $\Delta_i \leftarrow$ observations added to $\mathcal{H}_i$ since its last training call.
  if $n_i > 0$, $\Delta_i > 0$, and $i \bmod 2 = t \bmod 2$ **then**
    Let $P_i \leftarrow \lvert \mathbf{p}_i\rvert$ and $b_i \leftarrow 0$ if $P_i \le W_p$, otherwise $b_i \leftarrow (\mathbf{p}_i)_{P_i - W_p}$; set $L_i \leftarrow \{b_i+1, \ldots, n_i\}$. If $\lvert L_i\rvert > M$, keep M spaced indices across L_i, including its first and last indices (or only its last index when $M = 1$).
    $\Xi_{i,\mathrm{tr}} \leftarrow (\Xi_i^t)_{\cdot L_i}$, $y_{i,\mathrm{tr}} \leftarrow (y_i^t)_{L_i}$.
    $E_i \leftarrow \max\{E_{\mathrm{steps}}, \lceil E_{\mathrm{init}}\Delta_i/n_i\rceil\}$.
    $(\Theta_i, \mathcal{O}_i) \leftarrow \textsc{TrainNN}(\Theta_i, \Xi_{i,\mathrm{tr}}, y_{i,\mathrm{tr}}, E_i, \eta_{\mathrm{lr}}, \mathcal{O}_i)$; $\Delta_i \leftarrow 0$.
$n_b \leftarrow n_b^t$, $\Delta_b \leftarrow$ broker observations since last training.
if $n_b > 0$ and $\Delta_b > 0$ **then**
  Let $P_b \leftarrow \lvert \mathbf{p}_b\rvert$ and $b_b \leftarrow 0$ if $P_b \le W_p$, otherwise $b_b \leftarrow (\mathbf{p}_b)_{P_b - W_p}$; set $L_b \leftarrow \{b_b+1, \ldots, n_b\}$. If $\lvert L_b\rvert > M$, keep M spaced indices across L_b, including its first and last indices (or only its last index when $M = 1$).
  $\Psi_{b,\mathrm{tr}} \leftarrow \left[\psi_b\Big((X_{b,1}^t)_{\cdot\ell}, (X_{b,2}^t)_{\cdot\ell}\Big)\right]_{\ell\in L_b}$.
  $y_{b,\mathrm{tr}} \leftarrow (y_b^t)_{L_b}$.
  $E_b \leftarrow \max\{E_{\mathrm{steps}}, \lceil E_{\mathrm{init}}\Delta_b/n_b\rceil\}$.
  $(\Theta_b, \mathcal{O}_b) \leftarrow \textsc{TrainNN}(\Theta_b, \Psi_{b,\mathrm{tr}}, y_{b,\mathrm{tr}}, E_b, \eta_{\mathrm{lr}}, \mathcal{O}_b)$; $\Delta_b \leftarrow 0$.

4 Initialization

Calibration draws random pairs (i_ℓ, j_ℓ) and sets

$$\bar{q}_{\text{cal}} = \frac{1}{10{,}000} \sum_{\ell=1}^{10{,}000} \{Q + f(x_{i_\ell}, x_{j_\ell})\},$$
$$r = f_r \, \bar{q}_{\text{cal}}, \qquad \phi = c_s = \lambda_c \, \bar{q}_{\text{cal}}.$$

Local variables for Initialize

Symbol	Dimension/domain	Definition
λ_c	scalar	Cost rate used to set ϕ and c_s.
f_r	scalar	Reservation coefficient; sets $r = f_r \, \bar{q}_{\text{cal}}$ (default 0.60).
(i_ℓ, j_ℓ)	pair in $[N]^2$	Random calibration pair.
d_γ	integer	Number of active curve dimensions.
p_{rewire}	scalar	Watts-Strogatz rewiring probability.
ν_ℓ	integer	Frequency for active curve coordinate ℓ.
ψ_ℓ	scalar	Phase for active curve coordinate ℓ.
A_0	$\mathbb{R}^{d\times d}$	Raw matrix used to construct A.
Σ_x	$\mathbb{R}^{d\times d}$	Empirical agent type second moment.
B_0	$\mathbb{R}^{d\times d}$	Raw zero-trace symmetric regime matrix.
B_{raw}	$\mathbb{R}^{d\times d}$	Orthogonalized pre-normalization regime matrix.
$\Psi_{b,\text{seed}}, y_{b,\text{seed}}$	$\mathbb{R}^{d_b \times n_b^0}$, $\mathbb{R}^{n_b^0}$	Broker seed training batch using symmetric pair features.

Local variables for InitializeAgent

Symbol	Dimension/domain	Definition
σ_x	scalar	Agent type noise scale.
u_i	scalar	Agent i curve position.
ϵ_i	$\mathbb{R}^d$	Agent type noise for agent i.
π_{ij}^{ent}	scalar	Entrant attachment probability.
$\mathcal{V}_i^{\text{ent}}$	subset of $[N] \setminus \{i\}$	Entrant attachment set.

Algorithm 4 InitializeAgent

Require: State S, agent slot i, period counter t, parameters ϑ.
Ensure: Agent slot i is initialized in S.

1: Draw $u_i \sim \text{Unif}[0,1]$ and $\epsilon_i \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$.
2: $x_i^t \leftarrow (\gamma(u_i) + \epsilon_i)/\|\gamma(u_i) + \epsilon_i\|_2$.
3: Reset $\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$ to zero observations, $\bar{q}_{i\cdot}^t$, $C_{i\cdot}^t$, χ_i^t, M_i^t, and Δ_i^t. Draw W_{1i}^t with He initialization and set $b_{1i}^t = 0$, $w_{2i}^t = 0$, $b_{2i}^t = Q$; reset $\mathcal{O}_i^t$ to zero.
4: **if** $t = 0$ **then**
5: $\quad s_i^t \leftarrow (0, 0)$.
6: **else**
7: $\quad$ Remove i from Roster^t, D^t, and M_j^t for all j; delete all incident edges in G^t; set $\bar{q}_{ji}^t, C_{ji}^t \leftarrow 0$ for all $j \neq i$.

Sample $\mathcal{V}_i^{\text{ent}} \subset [N] \setminus \{i\}$ without replacement, $|\mathcal{V}_i^{\text{ent}}| = \min(\lfloor k_G/2 \rfloor, N-1)$, with

$$\pi_{ij}^{\text{ent}} \propto \exp\{-\|x_i^t - x_j^t\|_2^2\}.$$

Add edges (i, j) to G^t for all $j \in \mathcal{V}_i^{\text{ent}}$.

$s_{i,\text{self}}^t \leftarrow \begin{cases} |\mathcal{V}_i^{\text{ent}}|^{-1} \sum_{j \in \mathcal{V}_i^{\text{ent}}} s_{j,\text{self}}^t, & \mathcal{V}_i^{\text{ent}} \neq \emptyset, \\ 0, & \mathcal{V}_i^{\text{ent}} = \emptyset. \end{cases}$

$s_{i,\text{broker}}^t \leftarrow \text{rep}_b^t$.

Set $\mathbf{p}_i^t \leftarrow (0)$, treating entrant initialization as period 0 of its tenure.

Algorithm 5 Initialize

Require: Parameters ϑ and seed.

Ensure: Initial state S^0.

Draw curve frequencies $\nu_\ell \sim \text{Unif}\{1, \ldots, 5\}$ and phases $\psi_\ell \sim \text{Unif}[0, 2\pi)$ for $\ell = 1, \ldots, d_\gamma$, defining γ.

Allocate empty agent and broker containers in S^0 and store γ.

for $i = 1, \ldots, N$ **do**

  InitializeAgent$(S^0, i, 0, \vartheta)$.

Draw $u_c \sim \text{Unif}[0, 1]$ and $\epsilon_c \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$; set $c \leftarrow \gamma(u_c) + \epsilon_c$ without re-normalizing.

Draw $A_0 \in \mathbb{R}^{d \times d}$ iid standard normal and set $A \leftarrow A_0^\top A_0 \cdot d / \operatorname{tr}(A_0^\top A_0)$.

$\Sigma_x \leftarrow N^{-1} \sum_i x_i^0 (x_i^0)^\top$.

Draw symmetric zero-trace B_0 and set

$$B_{\text{raw}} \leftarrow B_0 - \frac{\operatorname{tr}(\Sigma_x B_0 \Sigma_x A)}{\operatorname{tr}(\Sigma_x A \Sigma_x A)} A, \qquad B \leftarrow B_{\text{raw}} / \|B_{\text{raw}}\|_F.$$

Calibrate $(\bar{q}_{\text{cal}}, r, \phi, c_s)$.

Build G^0 as a Watts-Strogatz graph on $[N]$ with degree k_G and rewiring probability p_{rewire}; add isolated broker node $N+1$.

Initialize $\mathcal{H}_b^0$ and set W_{1b}^0 using He initialization, $b_{1b}^0 = 0$, $w_{2b}^0 = 0$, $b_{2b}^0 = Q$, $\mathcal{O}_b^0 = 0$, and $\text{rep}_b^0 = 0$.

Sample $\text{Roster}^0 \subset [N]$ uniformly with $|\text{Roster}^0| = \lceil \alpha_R N \rceil$; add broker edges $(i, N+1)$ for all roster members.

for each unordered non-broker edge $\{i, j\} \in \mathcal{E}_G^0$ **do**

  Draw q_{ij} once; append (x_j^0, q_{ij}) to $\mathcal{H}_i^0$ and (x_i^0, q_{ij}) to $\mathcal{H}_j^0$; update $\bar{q}_{ij}^0, C_{ij}^0, \bar{q}_{ji}^0, C_{ji}^0$.

Let $\mathcal{B}_{\text{seed}} \leftarrow \{\{i, j\} \in \mathcal{E}_G^0 : i, j \in \text{Roster}^0\}$; shuffle and keep at most 100 edges.

for each $\{i, j\} \in \mathcal{B}_{\text{seed}}$ **do**

  Append (x_i^0, x_j^0, q_{ij}) to $\mathcal{H}_b^0$; increment n_b^0.

if $n_b^0 > 0$ **then**

  Set rep_b^0 to mean broker seed output.

else

  Set $\text{rep}_b^0 \leftarrow 0$.

Set $s_{i,\text{self}}^0$ to mean own seed output and $s_{i,\text{broker}}^0 \leftarrow \text{rep}_b^0$.

Set $\mathbf{p}_i^0 \leftarrow (n_i^0)$ for every agent and $\mathbf{p}_b^0 \leftarrow (n_b^0)$, so initialization is period 0; set $\Delta_i^0 \leftarrow n_i^0$ and $\Delta_b^0 \leftarrow n_b^0$.

for each i with $n_i^0 > 0$ **do**

  $(\Theta_i^0, \mathcal{O}_i^0) \leftarrow \text{TrainNN}(\Theta_i^0, \Xi_i^0, y_i^0, E_{\text{init}}, \eta_{\text{lr}}, \mathcal{O}_i^0)$; $\Delta_i^0 \leftarrow 0$.

if $n_b^0 > 0$ **then**

$\Psi_{b,\text{seed}} \leftarrow \Big[\psi_b\Big((X^0_{b,1})_{\cdot\ell}, (X^0_{b,2})_{\cdot\ell}\Big)\Big]_{\ell=1}^{n_b^0}$, $y_{b,\text{seed}} \leftarrow y_b^0$.

$(\Theta_b^0, \mathcal{O}_b^0) \leftarrow \textsc{TrainNN}(\Theta_b^0, \Psi_{b,\text{seed}}, y_{b,\text{seed}}, E_{\text{init}}, \eta_{\text{lr}}, \mathcal{O}_b^0)$; $\Delta_b^0 \leftarrow 0$.

return S^0.

5 Channel Choice

Local variables for ChooseChannel		
Symbol	Dimension/domain	Definition
U_{self}	scalar	Self-channel score.
U_{broker}	scalar	Broker-channel score.

Algorithm 6 ChooseChannel

Require: Agent i state (s_i, χ_i), broker reputation rep_b, RNG.

$U_{\text{self}} \leftarrow s_{i,\text{self}}$.

$U_{\text{broker}} \leftarrow \chi_i s_{i,\text{broker}} + (1-\chi_i)\text{rep}_b$.

if $U_{\text{self}} > U_{\text{broker}}$ **then**

  return self.

if $U_{\text{broker}} > U_{\text{self}}$ **then**

  return broker.

return self w.p. 1/2, else broker ▷ $U_{\text{self}} = U_{\text{broker}}$.

6 One Period

Local variables for PeriodUpdate		
Symbol	Dimension/domain	Definition
K	integer	Active-demand bound.
p_{demand}	scalar	Per-position demand probability.
α_R	scalar	Standing broker roster share.
p_{roster}	scalar	Roster-churn probability.
η_{exit}	scalar	Agent exit probability.
Δ_{net}	integer	Network-measure interval.

Algorithm 7 PeriodUpdate

Require: Previous state S^{t-1}, period t, parameters ϑ.

Ensure: Updated state S^t, accepted records $\mathcal{R}^t$, and pre-turnover measure row Y^t.

Clear accumulators, set $M_i^t \leftarrow \emptyset$ for all i, and set $D^t \leftarrow \emptyset$.

Refresh roster: remove each roster member with probability p_{roster}, then replenish uniformly to size $\lceil \alpha_R N \rceil$.

$\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t\}$.

$\mathcal{D}^t \leftarrow \emptyset$.

for $i = 1, \ldots, N$ **do**

    Draw $d_i^t \sim \text{Binomial}(K, p_{\text{demand}})$.
    if $d_i^t > 0$ **then**
        $z_i^t \leftarrow$ ChooseChannel(i, rep_b^{t-1}).
        Append (i, z_i^t, d_i^t) to $\mathcal{D}^t$.
        if $z_i^t =$ broker **then**
            Add i to D^t.
$\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t \cup D^t\}$.
TrainPredictors(S, t, ϑ).
$\mathcal{R}^t \leftarrow$ OfferMarket$(S, \mathcal{D}^t, \vartheta)$.
Synchronize broker-node edges to Roster^t, D^t, and current broker-channel matches.
UpdateSatisfaction$(S, \mathcal{D}^t, \mathcal{R}^t)$.
if $D^t \neq \emptyset$ **then**
    $\text{rep}_b^t \leftarrow |D^t|^{-1} \sum_{i \in D^t} s_{i,\text{broker}}^t$.
else
    $\text{rep}_b^t \leftarrow \text{rep}_b^{t-1}$.
if $t \bmod \Delta_{\text{net}} = 0$ **then**
    Recompute broker betweenness, Burt constraint, and effective size on the post-matching, pre-turnover graph G^t.
$Y^t \leftarrow$ CollectMeasures$(S^t, \mathcal{R}^t, \vartheta)$ on the post-matching, pre-turnover state.
Append n_i^t to $\mathbf{p}_i^t$ for every agent and append n_b^t to $\mathbf{p}_b^t$, recording the end-of-period training-window boundaries.
for $i = 1, \ldots, N$ **do**
    if agent i exits with probability η_{exit} **then**
        InitializeAgent(S^t, i, t, ϑ).
Synchronize broker-node edges to the post-turnover roster, clients, and active matches.
return $(S^t, \mathcal{R}^t, Y^t)$.

7 Offer Market

Local variables for OfferMarket

Symbol	Dimension/domain	Definition
$\hat{q}_i^t$	$\mathbb{R}^d \to \mathbb{R}$	Agent predictor.
$\hat{q}_b^t$	$\mathbb{R}^{d_b} \to \mathbb{R}$	Broker predictor.
$\mathcal{N}_G^t(i)$	subset of $[N]$	Non-broker graph neighbors of i.
U_i^t	subset of $[N]$	Self-search stranger pool independently sampled for demander i.
$n_{\text{strangers}}$	integer	Per-demander stranger-pool size.
$v_{i \to j}^{\text{self},t}$	scalar	Self-offer score.
Ω_i	integer	Residual offer quota for demander i.
Ω	$\mathbb{Z}_{\geq 0}^N$	Residual quota vector.
O^t	map	Directed offer book.
O_{ij}^t	offer record	Record (i, j, z, v) for offer $i \to j$.
$\mathcal{P}_O^t$	set	Unordered pairs represented in O^t.
$\mathcal{U}_i^{\text{self}}$	subset of $[N]$	Self-search candidate set.
$\mathcal{I}_i^{\text{self}}$	list	Self-search candidates sorted by score.
$\mathcal{P}_b^t$	set	Feasible unordered broker-pair set.

Local variables for OfferMarket (continued)

Symbol	Dimension/domain	Definition
$\mathcal{I}_{b,i}$	list	Accessible broker candidates for client i, sorted by v_b^t.
$\mathcal{B}_b^t$	list	Selected directed broker offers.
$u_{i\to j}^{\text{self}}, u_{b,i\to j}^{\text{local}}, u_{b,i\to j}^{\text{global}}$	$[0,1]$	Independent random secondary keys used only within equal-score rankings.
o, o', o_{ij}, o_{ji}	offer records	Offer records; fields include $o.z$ and $o.\text{from}$.
$z^\star$	channel	Accepted-record channel.
α, β	indices	Ordered endpoints induced by one unordered broker pair.
ι	$\{0,1\}$	Return value of AddOffer.

The symmetric broker score is

$$v_b^t(i,j) = \hat{q}_b^t\big(\psi_b(x_i^t, x_j^t)\big)\,.$$

Let $\mathcal{N}_G^t(i) = \{j \in [N] : (i,j) \in \mathcal{E}_G^t\}$ exclude the broker node. The self-search proposal score is defined for every graph neighbor and for the non-neighbors in the demander-specific stranger pool:

$$v_{i\to j}^{\text{self},t} = \begin{cases} \bar{q}_{ij}^t, & j \in \mathcal{N}_G^t(i),\ C_{ij}^t > 0, \\ \hat{q}_i^t(x_j^t), & j \in \mathcal{N}_G^t(i),\ C_{ij}^t = 0, \\ \hat{q}_i^t(x_j^t), & j \in U_i^t \setminus \mathcal{N}_G^t(i), \\ \perp, & \text{otherwise.} \end{cases}$$

Thus direct pair means replace predictor extrapolations when pair-specific experience exists; $\perp$ denotes no valid self-search score. The receiver-side acceptance score for a one-sided offer is

$$w_{j\leftarrow i}^t = \begin{cases} \bar{q}_{ji}^t, & (i,j) \in \mathcal{E}_G^t,\ C_{ji}^t > 0, \\ \hat{q}_j^t(x_i^t), & \text{otherwise.} \end{cases}$$

The market creates at most d_i^t outgoing offers from demander i. Receivers have no per-period capacity constraint; acceptance is pairwise, so a receiver may accept multiple offers in the same period. A demanded position can remain unmatched if no acceptable offer is formed.

Algorithm 8 AddOffer

Require: Offer book O^t, unordered-pair set $\mathcal{P}_O^t$, offer (i,j,z,v).

if O_{ij}^t is defined **then**
  return 0.
$O_{ij}^t \leftarrow (i,j,z,v)$.
if O_{ji}^t is undefined **then**
  $\mathcal{P}_O^t \leftarrow \mathcal{P}_O^t \cup \{\{i,j\}\}$.
return 1.

Algorithm 9 OfferMarket

Require: State S, demand records $\mathcal{D}^t$, parameters ϑ.
Ensure: Accepted records $\mathcal{R}^t$.

For all $(i, z_i, d_i) \in \mathcal{D}^t$, set residual offer quota $\Omega_i \leftarrow d_i$.

$O^t \leftarrow \emptyset$, $\mathcal{P}_O^t \leftarrow \emptyset$, $\mathcal{R}^t \leftarrow \emptyset$.
for each $(i, \mathrm{self}, d_i) \in \mathcal{D}^t$ with $\Omega_i > 0$ **do**
  Draw $U_i^t \subset [N]$ uniformly without replacement, $|U_i^t| = \min(n_{\mathrm{strangers}}, N)$.
  $\mathcal{U}_i^{\mathrm{self}} \leftarrow \mathcal{N}_G^t(i) \cup (U_i^t \setminus \mathcal{N}_G^t(i))$.
  Remove from $\mathcal{U}_i^{\mathrm{self}}$ the elements of $\{i\} \cup M_i^t$.
  Draw independent $u_{i\to j}^{\mathrm{self}} \sim \mathrm{Unif}(0,1)$ for each feasible j.
  $\mathcal{I}_i^{\mathrm{self}} \leftarrow \mathrm{argsort}_{j \in \mathcal{U}_i^{\mathrm{self}}:\ v_{i\to j}^{\mathrm{self},t} > r}(-v_{i\to j}^{\mathrm{self},t}, u_{i\to j}^{\mathrm{self}})$.
  for the first $\min(\Omega_i, |\mathcal{I}_i^{\mathrm{self}}|)$ agents j in $\mathcal{I}_i^{\mathrm{self}}$ **do**
    $\textsc{AddOffer}(O^t, \mathcal{P}_O^t, i, j, \mathrm{self}, v_{i\to j}^{\mathrm{self},t})$.
$\mathcal{A}_b^t \leftarrow \mathrm{Roster}^t \cup D^t$.
$D_+^t \leftarrow \{i : (i, \mathrm{broker}, d_i) \in \mathcal{D}^t,\ \Omega_i > 0\}$.
$\mathcal{P}_b^t \leftarrow \{\{i,j\} : i \neq j,\ (i \in D_+^t, j \in \mathcal{A}_b^t) \text{ or } (j \in D_+^t, i \in \mathcal{A}_b^t)\}$.
$\mathcal{B}_b^t \leftarrow \emptyset$.
for each $i \in D_+^t$ **do**
  Draw independent $u_{b,i\to j}^{\mathrm{local}} \sim \mathrm{Unif}(0,1)$ for each feasible j.
  $\mathcal{I}_{b,i} \leftarrow \mathrm{argsort}_{j \in \mathcal{A}_b^t : \{i,j\} \in \mathcal{P}_b^t,\ v_b^t(i,j) > r}(-v_b^t(i,j), u_{b,i\to j}^{\mathrm{local}})$.
  for the first $\min(\Omega_i, |\mathcal{I}_{b,i}|)$ agents j in $\mathcal{I}_{b,i}$ **do**
    Append $(i, j, v_b^t(i,j))$ to $\mathcal{B}_b^t$.
Draw a new independent $u_{b,i\to j}^{\mathrm{global}} \sim \mathrm{Unif}(0,1)$ for each $(i,j,v) \in \mathcal{B}_b^t$.
Sort $\mathcal{B}_b^t$ lexicographically by $(-v, u_{b,i\to j}^{\mathrm{global}})$.
for each $(i,j,v) \in \mathcal{B}_b^t$ **do**
  if $\Omega_i > 0$ **then**
    $\iota \leftarrow \textsc{AddOffer}(O^t, \mathcal{P}_O^t, i, j, \mathrm{broker}, v)$.
    $\Omega_i \leftarrow \Omega_i - \iota$.
for each $\{i,j\} \in \mathcal{P}_O^t$ **do**
  $o_{ij} \leftarrow O_{ij}^t$ if defined, else $\emptyset$; $o_{ji} \leftarrow O_{ji}^t$ if defined, else $\emptyset$.
  if $o_{ij} \neq \emptyset$ and $o_{ji} \neq \emptyset$ **then**
    $\textsc{FinalizeMatch}(S, \mathcal{R}^t, o_{ij}, o_{ji})$.
  else if $o_{ij} \neq \emptyset$ and $w_{j\leftarrow i}^t > r$ **then**
    $\textsc{FinalizeMatch}(S, \mathcal{R}^t, o_{ij}, \emptyset)$.
  else if $o_{ji} \neq \emptyset$ and $w_{i\leftarrow j}^t > r$ **then**
    $\textsc{FinalizeMatch}(S, \mathcal{R}^t, o_{ji}, \emptyset)$.
return $\mathcal{R}^t$.

Algorithm 10 FinalizeMatch

Require: State S, accepted-record list $\mathcal{R}^t$, offer $o = (i, j, z, v)$, optional reverse offer o'.
if $j \in M_i^t$ **then**
  return .
Draw $q_{ij}^t = Q + f(x_i^t, x_j^t) + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$.
Append columns x_j^t to Ξ_i and x_i^t to Ξ_j; append q_{ij}^t to y_i and y_j; increment n_i and n_j.
Update $\bar{q}_{ij}, C_{ij}$ and $\bar{q}_{ji}, C_{ji}$.
Add edge (i,j) to G^t; add j to M_i^t and i to M_j^t.
if $z = \mathrm{broker}$ or ($o' \neq \emptyset$ and $o'.z = \mathrm{broker}$) **then**
  Append the two party types x_i^t and x_j^t to $X_{b,1}$ and $X_{b,2}$; append q_{ij}^t to y_b; increment n_b.
  $z^\star \leftarrow \mathrm{broker}$.

else
  $z^\star \leftarrow$ self.
Append accepted record $(i, j, z^\star, q^t_{ij}, v, o, o')$ to $\mathcal{R}^t$.

8 Satisfaction and Measures

Local variables for satisfaction

Symbol	Dimension/domain	Definition
Y_i	scalar	Agent i period satisfaction numerator.
$n_{i,b}$	integer	Accepted broker offers credited to demander i.
$\tilde{q}^t_i$	scalar	Channel-specific satisfaction input.
ζ	accepted record	Iterator over $\mathcal{R}^t$.
$\mathbf{1}\{\cdot\}$	indicator	Equals 1 when its condition is true and 0 otherwise.

Algorithm 11 UpdateSatisfaction

Require: State S, demand records $\mathcal{D}^t$, accepted records $\mathcal{R}^t$.
for each $(i, z_i, d_i) \in \mathcal{D}^t$ **do**
  $Y_i \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset,\ o.\text{from} = i\}\, \zeta.q.$
  $n_{i,b} \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset,\ o.\text{from} = i,\ o.z = \text{broker}\}.$
  if $z_i =$ self **then**
    $\tilde{q}^t_i \leftarrow Y_i/d_i - c_s.$
  else
    $\tilde{q}^t_i \leftarrow (Y_i - \phi n_{i,b})/d_i;\ \chi^t_i \leftarrow 1.$
  $s^t_{i,z_i} \leftarrow (1-\omega)s^{t-1}_{i,z_i} + \omega \tilde{q}^t_i.$

CollectMeasures. CollectMeasures$(S^t, \mathcal{R}^t, \vartheta)$ is a pure measure-collection function called after matching and before entry/exit turnover. It records demand, outsourcing, accepted matches, realized output, selected-sample prediction quality, deterministic holdout prediction quality, broker access and assessment counts, satisfaction, reputation, roster size, broker access size, agent-node degree summaries excluding the broker node, and cached broker network measures. The measures themselves do not feed back into S^{t+1}. For rank correlation, predicted values first produce a strict ranking: equal predictions are ordered uniformly at random, independently for the agent and broker, while equal realized outputs retain average ranks. Spearman correlation is computed only after this predicted ranking is formed. The same rule is used for each holdout candidate set and for pooled accepted directed offers within each selected-offer channel.